Unified Study: Single-Source Self-Healing Distribution Network and Distributed Edge FLISR

A combined project paper for a third-year EEE project, merging the project concept notes with the revision notes on arXiv:2407.17497v1 ("A Distributed Edge FLISR Solution & Network Simulation Test Platform")

1. Abstract

This paper combines two source documents into one project framework. The first is a planning document proposing a single-source, six-bus, three-phase distribution network with fault detection, isolation and restoration through a normally open tie point. The second is a set of study notes on a published paper describing a distributed edge FLISR (Fault Location, Isolation and Service Restoration) solution with a network simulation test platform. Both address the same engineering problem: a fault should not take down healthy customers when a safe alternative supply path exists. The unified project builds a small self-healing feeder, first in simulation and then as a low-voltage hardware demonstrator, and compares centralized against local (distributed) decision-making under communication delay and loss. Section 9 lists where the two sources differ in concept.

2. Problem Statement

Distribution networks consist of feeders, branches, switches, protection devices and loads. When a fault occurs, protection may disconnect a larger area than the faulted section, so healthy customers lose supply. Manual restoration (locate, dispatch a crew, operate switches) is slow. Automated FLISR reduces the number of customers affected and the duration of interruption.

The unified project asks: **can a small distribution network automatically detect a fault, identify the faulted section, isolate it at both ends, and restore healthy loads through an alternate path, and how does the choice of control architecture affect speed and reliability?**

3. Background Concepts Common to Both Sources

3.1 The FLISR sequence

Detect, locate, isolate, check the alternate path, restore healthy loads.

Step	Meaning	Typical action
Fault detection	Recognize an abnormal condition	Use current, voltage, protection status or an event flag
Location	Identify the likely faulted section	Combine measurements or indicators with known topology
Isolation	Disconnect the faulted section	Open the switches at both ends and verify their state
Service restoration	Re-energize healthy sections if safe	Close a suitable tie switch or reconfigure the network

3.2 Key terms

- **Bus:** a common electrical node. **Feeder section:** the line between two buses. Three-phase feeders carry phases A, B and C.
- **Normally open point (NOP) / tie switch:** an open switch that, when closed after isolation, gives healthy buses an alternate supply route.
- **FPI (Fault Passage Indicator):** indicates that fault current passed a monitored point. It is evidence, not an exact fault locator.
- **LVI (Loss of Voltage Indicator):** indicates lost voltage at a monitored point. Loss of voltage can also come from upstream switching or supply loss.
- **Topology data:** connectivity of sources, lines, switches and loads. Wrong topology data can produce unsafe restoration decisions.

3.3 Detection is not location

Both sources agree that sensing an abnormal condition is not the same as knowing which section is faulted. Location needs extra measurements or indicator states plus knowledge of the network topology.

3.4 Restoration is conditional

Neither source claims that every fault can be restored. Success depends on fault location, alternate paths, switch placement, load limits, source or feeder capacity, voltage limits, protection coordination and topology. Multiple faults leave more customers disconnected.

4. Control Architectures

4.1 Centralized control

Field devices send measurements and status to one controller, which evaluates the network and issues switching commands. It is simpler to design and debug and has a wide view of the network, but communication delay, communication loss or controller failure can delay or prevent action.

4.2 Decentralized / distributed (edge) control

Local controllers run decision logic close to the equipment. The planning notes describe one controller per bus or protection zone exchanging status with neighbours. The paper describes an edge controller that makes local FLISR decisions and still interacts with a central system when available. Benefits include less dependence on a round trip to a control centre; challenges include coordination, timing, inconsistent information and communication loss.

4.3 Key independence principle

The number of power sources does not decide whether control is centralized or decentralized. A single source can feed a network supervised by several cooperating local controllers.

5. Unified Project Concept

Working title: Design and Development of a Decentralized Fault Detection, Isolation and Self-Healing Distribution Network with Hardware-in-the-Loop Validation.

Core idea: one three-phase source feeds a six-bus network arranged as a ring but operated radially, with one NOP. The electrical behaviour is studied in simulation. A scaled low-voltage hardware model, controlled by an ESP32, demonstrates the switching logic. Faults are introduced virtually or in a controlled low-voltage way, never as destructive faults on laboratory AC equipment.

5.1 Two complementary layers

Layer	Purpose	Fidelity	Source of the idea
Simulation layer	Three-phase network behaviour, fault types (LG, LL, LLG, three-phase), voltage and current signatures, power flow, voltage-drop and capacity checks	High electrical realism	Planning notes
Hardware demonstrator	Sensing, ADC inputs, ESP32 logic, relay-driven switches, switch-state feedback, LED loads, display or dashboard	Functional logic, not AC power behaviour	FLISR paper notes (adaptation)

The two layers can be coupled: a MATLAB/Simulink or Python network model can supply measurements to the controller, or the controller can drive the model's switches.

5.2 Normal and fault-time operation

Normal path: Source, BUS 1, BUS 2, BUS 3, BUS 4, BUS 5, BUS 6, with the BUS 6 to BUS 1 tie open. After a fault in a section, the controller opens the breakers at both ends of that section, confirms they opened, checks that the alternate path has adequate capacity and acceptable voltage, then closes the tie so healthy buses are supplied from the opposite direction.

5.3 Physical set-up for the hardware demonstrator

- Isolated, current-limited low-voltage DC supply.
- Each feeder section represented by wiring, a controllable switch and a low-voltage load (LED with resistor or similar).
- Current and voltage sensing at selected points, with scaled and protected outputs to the ESP32 ADC.
- Relay modules or transistor drivers; never drive a relay coil directly from a GPIO.
- Switch-state feedback so the controller can verify that a command succeeded.
- LED load indicators plus a display or dashboard showing fault location and switching state.

This is a functional architecture, not a validated schematic or bill of materials. Ratings, wiring, ADC scaling and isolation must be designed for the chosen parts.

6. Fault Demonstration Method

Method	Description	Strength / limitation
Physical	Controlled abnormal condition on a low-voltage test feeder	Real sensor values, but few safe scenarios
Virtual	Fault injected as a change in simulated network or switch state	Many repeatable and rare cases, but does not prove real sensing
Hybrid	Real sensing and switching hardware with virtual faults or communication events	Best balance

Recommendation (shared by both sources' spirit): use hybrid testing and label every result as measured on hardware, simulated, or software-generated.

7. Development Stages

1. **Core electrical model.** Build and validate the single-source six-bus network; establish normal voltages, currents and power flow.
2. **Fault studies.** Simulate selected faults and record voltage/current signatures; decide how location is inferred.
3. **Isolation.** Implement two-ended isolation with interlocking and breaker-status checks.
4. **Restoration.** Close the tie after isolation; determine which healthy buses can be restored; verify feeder and source limits.
5. **Centralized baseline, then decentralized coordination.** Add local controllers and communication; compare against the centralized baseline.
6. **Hardware-in-the-loop demonstrator.** Implement the ESP32 logic with sensing, relays and feedback.
7. **Optional extension.** Add PV (for example at BUS 3) and battery storage (for example at BUS 5), bidirectional flows, local energy management and islanding. Genuine P2P energy trading only becomes possible at this stage, because passive loads have no surplus to share.

8. Experiments and Metrics

8.1 Experiments

#	Experiment	What to measure or compare
1	Baseline, no FLISR	Loads disconnected and for how long
2	Centralized FLISR	Restoration time, loads affected, communication dependence
3	Distributed FLISR	Local decision time and restoration success
4	Add communication delay	Increase in centralized response time
5	Lose communication	Whether local logic still operates safely
6	Multiple faults or unavailable tie	Which loads cannot be restored and why
7	Switch fails to operate	Whether feedback detects it and blocks unsafe follow-on actions

8.2 Metrics

- Fault-detection accuracy across the fault types and cases tested.
- Correct identification of the faulted section.
- Detection, isolation and restoration times.
- Loads interrupted, and number or percentage of healthy loads restored.
- Voltage profile and feeder current before and after reconfiguration.
- Performance under communication delay, packet loss or outage.
- Safety checks: no path may energize the faulted section or improperly parallel sources.

8.3 Reference results from the published paper

The paper reports results for its own test scenarios. They are examples, not targets or universal values.

Reported item	Before	After
Waterford scenario, customers affected	1,370	171 (about 87.5% fewer)
Customer-minutes lost, as reported per hour	78,090	9,747

Method	Average operation time reported
5G centralized	212.8 ms
4G LTE centralized	260.4 ms
3G centralized	267.1 ms
2G centralized	1,049.6 ms
Distributed FLISR	about 2 ms

The roughly 2 ms figure is a local algorithm time in the authors' setup. It may exclude sensor delay, breaker operating time, inter-device communication and safety checks, so it must not be presented as a general FLISR response time. Verify all figures against the original paper before using them in a formal report.

9. Differences in Concept Between the Two Sources

The sources are compatible, but they are not the same kind of document and they emphasize different things.

Aspect	Single-source project notes	Distributed edge FLISR paper notes
Nature of document	Original project proposal and plan	Study summary of a published paper, plus labelled adaptations
Main goal	Design a small network and its self-healing logic	Present an edge FLISR solution and a simulation test platform
Network model	Six-bus three-phase ring, operated radially with one NOP	Real utility-style topologies (Waterford, Kerry, Mullingar) and a generic feeder with sections and switches
Electrical treatment	AC three-phase, with LG, LL, LLG and three-phase faults and electrical signatures	Faults treated mainly as network and switch-state events; no electronics schematic in the paper
Detection basis	Voltage and current behaviour	Indicators (FPI, LVI), switch status and topology data

Aspect	Single-source project notes	Distributed edge FLISR paper notes
Control terminology	Decentralized or multi-agent controllers, one per bus or zone, exchanging neighbour information	Distributed edge controller at the equipment, linked to a central system when available
Starting point for control	Centralized baseline first, then decentralized	Compares centralized (over 2G to 5G links) with distributed edge
Role of communication	Neighbour-to-neighbour coordination; delay and loss tested	Central focus: latency and availability of cellular links versus local decisions
Industrial protocols	Not covered	RTU, MQTT and IEC 60870-5-104 appear; not required for a student build
Hardware	Low-voltage signals or simulated measurements, no destructive faults	ESP32-based low-voltage model is an adaptation, not from the paper
Fault demonstration	Simulated faults of several types	Physical, virtual or hybrid, with hybrid recommended
Results and metrics	Planned measurable outcomes, no results yet	Reported customer-count, customer-minutes and operation-time results
DER and P2P	Explicit optional extension (PV, battery, islanding); P2P distinguished from control architecture	Not part of the scope
Source-failure limitation	States that a single source failure cannot be survived without another source or storage	Not discussed in the notes

9.1 Points of agreement

- The FLISR sequence and the need for isolation before restoration.
- Restoration through an alternate path, constrained by capacity, voltage and safety.
- Verification of switch operation rather than trusting commands.
- Avoiding destructive tests; preferring simulation or low-voltage, hybrid testing.

- Comparing centralized and local decision-making.
- Honest limits: not every load can be restored after every fault.

9.2 Interpretation of the main conceptual difference

The planning notes treat control architecture as a **topology of cooperating controllers** (each bus or zone has an agent), whereas the paper treats it as **where the computation runs** (edge versus central system) and quantifies the effect of communication technology. The unified project uses both views: local controllers per zone (planning notes) justified and evaluated using communication-delay experiments (paper).

10. Engineering Cautions

- Do not assume closing the tie always works; the alternate path may be overloaded or have unacceptable voltage drop.
- Isolate the faulted section at both ends before connecting an alternate path.
- Use interlocking and status confirmation to avoid unintended source paralleling.
- A single source cannot maintain supply if it fails; this would need another source or storage.
- Never connect an experimental circuit to mains or a real feeder; use an isolated, current-limited low-voltage supply.
- Incorrect topology data can cause unsafe restoration, so validate the network model.

11. Decisions to Settle Before Detailed Design

1. Ring operated with one NOP, or radial feeder with a tie line?
2. Number of buses and voltage and power ratings in simulation (six buses suggested).
3. Fault types in the first demonstration.
4. Centralized controller first, then decentralized, as planned?
5. Communication method between controllers (for example wired serial, Wi-Fi, or MQTT-style messaging).
6. Simulation tool: MATLAB/Simulink or Python.
7. How faults will be injected into the hardware (virtual, physical or hybrid).

12. Conclusion

The single-source project notes give the electrical and staged-development backbone; the FLISR paper notes supply the control-architecture comparison, the indicator-based fault logic, the hybrid test philosophy and reference metrics. Combined, they define a

focused and defensible project: a six-bus self-healing feeder with centralized and distributed control, validated in simulation and a low-voltage ESP32 hardware demonstrator, with PV, storage and P2P trading kept as later extensions.

13. References

- Single-Source Self-Healing Distribution Network: project concept notes (supplied document).
- Distributed Edge FLISR: revision notes on the paper (supplied document).
- *A Distributed Edge FLISR Solution & Network Simulation Test Platform*, arXiv:2407.17497v1, https://arxiv.org/html/2407.17497v1